

3.2.2 Variances of U-statistics

If $E[h(X_1, \dots, X_m)]^2 < \infty$, then the variance of U_n in (3.11) with kernel h has an explicit form. To derive $\text{Var}(U_n)$, we need some notation. For $k = 1, \dots, m$, let

$$\begin{aligned} h_k(x_1, \dots, x_k) &= E[h(X_1, \dots, X_m) | X_1 = x_1, \dots, X_k = x_k] \\ &= E[h(x_1, \dots, x_k, X_{k+1}, \dots, X_m)]. \end{aligned}$$

Note that $h_m = h$. It can be shown that

$$h_k(x_1, \dots, x_k) = E[h_{k+1}(x_1, \dots, x_k, X_{k+1})]. \quad (3.14)$$

Define

$$\tilde{h}_k = h_k - E[h(X_1, \dots, X_m)], \quad (3.15)$$

$k = 1, \dots, m$, and $\tilde{h} = \tilde{h}_m$. Then, for any U_n defined by (3.11),

$$U_n - E(U_n) = \binom{n}{m}^{-1} \sum_c \tilde{h}(X_{i_1}, \dots, X_{i_m}). \quad (3.16)$$

Theorem 3.4 (Hoeffding's theorem). For a U-statistic U_n given by (3.11) with $E[h(X_1, \dots, X_m)]^2 < \infty$,

$$\text{Var}(U_n) = \binom{n}{m}^{-1} \sum_{k=1}^m \binom{m}{k} \binom{n-m}{m-k} \zeta_k,$$

where

$$\zeta_k = \text{Var}(h_k(X_1, \dots, X_k)).$$

Proof. Consider two sets $\{i_1, \dots, i_m\}$ and $\{j_1, \dots, j_m\}$ of m distinct integers from $\{1, \dots, n\}$ with exactly k integers in common. The number of distinct choices of two such sets is $\binom{n}{m} \binom{m}{k} \binom{n-m}{m-k}$. By the symmetry of $\tilde{h}_m$ and independence of $X_1, \dots, X_n$,

$$E[\tilde{h}(X_{i_1}, \dots, X_{i_m}) \tilde{h}(X_{j_1}, \dots, X_{j_m})] = \zeta_k \quad (3.17)$$

for $k = 1, \dots, m$ (exercise). Then, by (3.16),

$$\begin{aligned} \text{Var}(U_n) &= \binom{n}{m}^{-2} \sum_c \sum_c E[\tilde{h}(X_{i_1}, \dots, X_{i_m}) \tilde{h}(X_{j_1}, \dots, X_{j_m})] \\ &= \binom{n}{m}^{-2} \sum_{k=1}^m \binom{n}{m} \binom{m}{k} \binom{n-m}{m-k} \zeta_k. \end{aligned}$$

This proves the result. ■